APEX INTERNATIONAL ACADEMY

CBSE Senior Secondary Affiliation No. 1930482 | Academic Excellence & Research Division

OFFICIAL ACADEMIC SOLUTION & EVALUATION DOSSIER

VERIFIED & GRADED

ASSIGNMENT: Algebraic Expressions & Linear Equations

SUBJECT: Mathematics (MATH101)

CLASS & SECTION: Class 7 - Section A

ACADEMIC YEAR: 2025 - 2026

MARKS EVALUATED: 88 / 100 (Grade A)

TEACHER DIRECTIVE & PROBLEM FOCUS:
Solve practice exercise 4.2 focusing on variables, constants, balancing methods, and linear equations in one variable.

1. Executive Overview & Problem Context

Algebra generalizes arithmetic through the introduction of symbolic variables representing unknown quantities. A linear equation in one variable expresses an equality between two algebraic expressions of the first degree (ax + b = c, where a "` 0), solvable via systematic inverse operations and transposition.

2. Core Analysis, Solutions & Systematic Working

- **Variables, Coefficients & Polynomial Terminology:**

  In $7x^2 - 5x + 12$: x is the variable, 7 is the coefficient of x^2, -5 is the coefficient of x, and 12 is the numerical constant. Like terms share identical variable factors with equal exponents and can be combined algebraically.

- **Balance Method vs Transposition Rules:**

  An equation acts as a balanced scale: operations applied to the LHS must be simultaneously applied to the RHS. Transposition shifts terms across the equality sign with sign inversion: addition becomes subtraction, multiplication becomes division.

- **Worked Example 1: Multi-Step Linear Equation:**

  Solve: 5(2x - 3) - 3(x + 4) = 22. Expand brackets: 10x - 15 - 3x - 12 = 22. Combine like terms: 7x - 27 = 22. Transpose 27: 7x = 22 + 27 = 49. Divide by 7: x = 49 / 7 = 7. Verification: LHS = 5(11) - 3(11) = 55 - 33 = 22 = RHS.

- **Worked Example 2: Fractional Coefficients:**

  Solve: (3x/4) - (x/2) = 5. Find common denominator (4): (3x - 2x)/4 = 5 -> x/4 = 5 -> x = 20. Verification confirms LHS = 15 - 10 = 5.

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3. Detailed Data Matrix & Systematic Findings

Equation	Method Applied	Intermediate Step	Solution (x)	Verification (LHS = RHS)
3x - 7 = 14	Transposition of -7	3x = 21	x = 7	3(7) - 7 = 14 (True)
4(x + 2) = 28	Division by 4	x + 2 = 7	x = 5	4(5 + 2) = 28 (True)
2x + 9 = 5x - 6	Variable Isolation	3x = 15	x = 5	10 + 9 = 25 - 6 = 19 (True)
(x / 3) + 4 = 9	Transposition of +4	x / 3 = 5	x = 15	(15/3) + 4 = 9 (True)
7x - (2x + 5) = 20	Bracket Expansion	5x - 5 = 20 -> 5x = 25	x = 5	35 - 15 = 20 (True)

4. Student Conclusions & Reflective Synthesis

Linear equations provide the foundational mathematical framework for expressing conservation laws, rate-time-distance dynamics, and economic equilibrium models in scientific disciplines.

5. Official Faculty Evaluation & Grading Rubric

ASSESSMENT CRITERIA BREAKDOWN

Conceptual Clarity & Factual Accuracy: 28 / 30

Problem Solving & Structured Working: 18 / 20

Analytical Depth & Research Methodology: 28 / 30

Presentation, Organization & Neatness: 14 / 20

FACULTY FEEDBACK REMARK:

"Very good effort. Clear and organized transposition steps. Word problem formulation is logically structured."

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